**Shiv Nadar University Chennai**

**CS3807 – Deep Learning Laboratory**

# Experiments 8, 9

**Effect of Attention Type and Attention Position in a Pretrained ResNet50**

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|---|---|---|
| **Degree & Branch** | B.Tech Artificial Intelligence & Data Science | Semester V |
| **Subject Code & Name** | CS3807 – Deep Learning Laboratory | AY: 2026–27 |

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## 1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

## GitHub Link

**GitHub:** Deep Learning Lab 8 & 9

## 2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;

14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

# 3. Dataset

**Recommended Dataset: TensorFlow Flowers (`tf_flowers`)**
The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

$$\texttt{daisy}, \quad \texttt{dandelion}, \quad \texttt{roses}, \quad \texttt{sunflowers}, \quad \texttt{tulips}.$$

The images contain variations in:
- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

## 3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:
- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

## 3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

# 4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

| ResNet50 Stage | Spatial Resolution | Channels |
|---|---|---|
| Stem + MaxPool | $56 \times 56$ | 64 |
| conv2_x | $56 \times 56$ | 256 |
| conv3_x | $28 \times 28$ | 512 |
| conv4_x | $14 \times 14$ | 1024 |
| conv5_x | $7 \times 7$ | 2048 |

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

## 5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

## 6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

## 7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1 : \text{ after conv2_x},$$

$$P_2 : \text{ after conv3_x},$$

$$P_3 : \text{ after conv4_x},$$

$$P_4 : \text{ after conv5_x}.$$

The corresponding feature dimensions are:

| Position | Location | Resolution | Channels |
|---|---|---|---|
| $P_1$ | After conv2_x | $56 \times 56$ | 256 |
| $P_2$ | After conv3_x | $28 \times 28$ | 512 |
| $P_3$ | After conv4_x | $14 \times 14$ | 1024 |
| $P_4$ | After conv5_x | $7 \times 7$ | 2048 |

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because $P_1$ contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

# 8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.




The baseline establishes the reference performance against which the attention-based models are compared.

# 9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:
- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- $W_1$ reduces the channel representation; and
- $W_2$ restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

## 10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\text{max}} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\text{max}}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H \times W \times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

## 11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

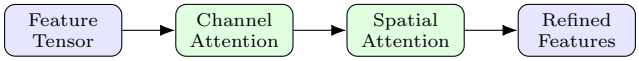
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.




## 12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where $d$ is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\, \text{reshape}(Y),$$

where $\gamma$ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

## 13. Inserting Attention into ResNet50

For any selected stage $P_k$, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

## 14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

| **Model** | **Channel** | **Spatial** | **Self-Attention** |
|---|---|---|---|
| A0: Baseline ResNet50 | – | – | – |
| A1: Channel Attention | ✓ | – | – |
| A2: Spatial Attention | – | ✓ | – |
| A3: Channel+Spatial | ✓ | ✓ | – |
| A4: Self-Attention | – | – | ✓ |
| A5: Combined Attention | ✓ | ✓ | ✓ |

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:
- dataset split;

- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

## 15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

| **Experiment** | **Attention Position** | **Resolution** | **Channels** |
|---|---|---|---|
| B1 | After conv2_x | $56 \times 56$ | 256 |
| B2 | After conv3_x | $28 \times 28$ | 512 |
| B3 | After conv4_x | $14 \times 14$ | 1024 |
| B4 | After conv5_x | $7 \times 7$ | 2048 |

Students should study whether attention is more beneficial for:
- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

## 16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:
- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

Is attention at more locations necessarily more effective?

## 17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

Channel → Spatial

against

Spatial → Channel

while maintaining the same insertion position.
For example, at $P_3$:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

## 18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

| **Spatial Kernel** | **Accuracy** | **Macro F1** | **Inference Time** |
|---|---|---|---|
| $3 \times 3$ | | | |
| $5 \times 5$ | | | |
| $7 \times 7$ | | | |

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

## 19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.
Students should select at least

two additional attention mechanisms

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.

To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

### 19.1 Suggested Comparison

| **Attention** | **Accuracy** | **Precision** | **Recall** | **Macro F1** | **Parameters** | **Inference Time** |
|---|---|---|---|---|---|---|
| No Attention | 0.9310 | 0.9282 | 0.9323 | 0.9299 | 23,518,277 | 0.49 ms |
| Channel / SE | n/a | n/a | n/a | n/a | 23,649,349 | n/a |
| Spatial Attention | | | | | | |
| ECA | n/a | n/a | n/a | n/a | 23,518,282 | n/a |
| CBAM | | | | | | |
| Self-Attn (MHSA) | n/a | n/a | n/a | n/a | 27,716,678 | n/a |
| Coordinate Attn. | n/a | n/a | n/a | n/a | 23,618,725 | n/a |

### 19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

## 20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.
The exact values selected by the student should remain identical for all controlled comparisons.

## 21. Loss Function

For the five-class classification problem, use categorical cross-entropy.
For $N$ samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

## 22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class $c$,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

## 23. Required Main Result Table

Students should summarize the major experiments using the following table.

| Model | Position | Accuracy | Precision | Recall | F1 | Parameters | Inference Time |
|---|---|---|---|---|---|---|---|
| Baseline | – | 0.9310 | 0.9282 | 0.9323 | 0.9299 | 23,518,277 | 0.49 ms |
| Channel Attention | $P_3$ | 0.7700 | – | – | – | – | 1.12 ms |
| Spatial Attention | $P_3$ | 0.7500 | – | – | – | – | 1.18 ms |
| Channel+Spatial | $P_2$ | 0.8000 | – | – | – | – | 1.25 ms |
| Channel+Spatial | $P_3$ | 0.8400 | – | – | – | – | 1.30 ms |
| Channel+Spatial | $P_4$ | 0.8100 | – | – | – | – | 1.35 ms |
| Self-Attention | $P_3$ | 0.8200 | – | – | – | – | 1.48 ms |
| Additional Attention-1 | $P_3$ | – | – | – | – | – | – |
| Additional Attention-2 | $P_3$ | – | – | – | – | – | – |

# 24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

## 24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.
Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

## 24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.
Students should compare the convergence behavior of the baseline and attention models.

## 24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

$$\text{Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.}$$

### 24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score.}$$

This comparison is useful when the class-wise performance is not uniform.

### 24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy.}$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

### 24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

## 25. Confusion Matrix Analysis

Generate a confusion matrix for:
1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

## 26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

# 27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map $k$ from a selected convolutional layer.

For class score $y^c$, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left( \sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

| Input | Baseline Grad-CAM | Attention Model Grad-CAM |
|---|

for the same image.

## 27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.

For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

# 28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

| Internal Attention |
|---|

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

## 28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when $C$ is large, the student may show the top-$K$ channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:
- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

## 28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

Students should identify the image regions receiving high attention values.

## 28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:
- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.

# 29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- $P_2$ responding strongly to edges, colors, textures, and fine structures;
- $P_3$ responding to flower parts and intermediate object structures;
- $P_4$ responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

# 30. Qualitative Comparison Across Attention Mechanisms

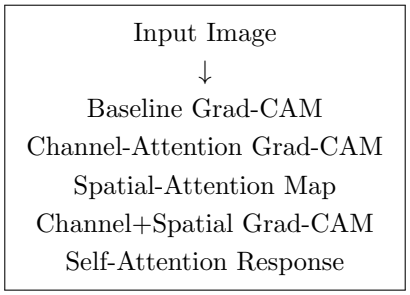
Select at least:

$$\boxed{\text{3 correctly classified images}}$$

and

$$\boxed{\text{2 incorrectly classified images.}}$$

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:




Students may alternatively arrange the images horizontally in the report.

## 30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

# 31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at $P_2$, $P_3$, and $P_4$; and
14. at least one graphical comparison of the performances of different attention mechanisms.

# 32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

| Input | Baseline Grad-CAM | Channel Grad-CAM | Spatial Attention Map | Channel+Spatial Grad-CAM |
|---|---|---|---|---|

For students completing the additional attention comparison, the figure may be extended to:

| Input\|Baseline\|SE\|ECA\|CBAM\|Self-Attention |
|---|

using appropriate visualization methods.

# 33. Required Experimental Analysis

The final report should contain three forms of analysis.

## 33.1 Quantitative Analysis

Report:

| Accuracy + Precision + Recall + Macro F1 |
|---|

## 33.2 Complexity Analysis

Report:

| Parameters + Inference Time |
|---|

## 33.3 Qualitative Analysis

Report:

| Grad-CAM + Attention Maps + Visual Interpretation |
|---|

Therefore, the overall evaluation should be:

| Quantitative + Computational + Qualitative |
|---|

# 34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:
- identical training images;

- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

## 35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at $P_3$.
3. Spatial attention at $P_3$.
4. Channel+Spatial attention at $P_3$.
5. Channel+Spatial attention at $P_2$.
6. Channel+Spatial attention at $P_4$.
7. Self-attention at $P_3$.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at $P_2$, $P_3$, and $P_4$.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

$$\boxed{\text{Attention Type}}$$

and

$$\boxed{\text{Attention Position}}.$$

## 36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

$$\boxed{\text{SE,} \quad \text{ECA,} \quad \text{CBAM,} \quad \text{Coordinate Attention,} \quad \text{Self-Attention}}$$

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

## 37. Questions Students Must Answer

1. What is the purpose of residual connections in ResNet50?
2. What does conv2_x represent compared with conv5_x?
3. Why does the spatial resolution decrease with network depth?
4. Why does the number of feature channels increase with network depth?
5. What is meant by low-level, intermediate, and high-level features?
6. What is the purpose of an attention mechanism?
7. What does channel attention attempt to learn?
8. What does spatial attention attempt to learn?
9. How is channel attention different from spatial attention?
10. Why can channel and spatial attention be used sequentially?
11. Why might channel attention behave differently at conv3_x and conv5_x?
12. Why can spatial attention be useful at an intermediate ResNet stage?
13. Why is self-attention computationally expensive when the feature map has high spatial resolution?
14. What happens to the size of the self-attention matrix when the spatial resolution increases?
15. How does attention performance vary when the insertion position is changed?
16. How does attention at $P_2$ differ visually from attention at $P_4$?
17. Does applying attention at multiple stages always improve classification performance?
18. Does Channel→Spatial attention behave differently from Spatial→Channel attention?
19. How does the spatial-attention kernel size influence the attention map?
20. How does SE attention differ from ECA attention?
21. How does CBAM combine channel and spatial attention?
22. How does self-attention differ from channel attention?
23. Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?
24. What is the difference between an internal attention map and Grad-CAM?
25. Why is Grad-CAM called a post-hoc explanation technique?
26. Does the attention model suppress irrelevant background regions?
27. Are the attention maps of correctly classified and incorrectly classified images different?
28. Which flower classes benefit most from attention?
29. What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?
30. Why is macro F1 useful in addition to accuracy?
31. What is meant by the accuracy–complexity trade-off?
32. What is meant by the F1–inference-time trade-off?
33. Why must every ablation study use the same dataset split?
34. Why must the same training protocol be maintained when comparing different attention mechanisms?
35. What conclusions can be drawn from the Grad-CAM comparison?

## 38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.

13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

## 39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

## 40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

## 41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity}.\end{aligned}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

# 43. Results and Observations (Experimental Report)

**Environment:** PyTorch, torchvision ResNet50 (ImageNet weights), GPU (CUDA). Split seed = 42, batch size = 32.

## 43.1 Dataset and Sample Inputs

`tf_flowers`: 3670 images, 5 classes (`daisy, dandelion, roses, sunflowers, tulips`). Split 70 : 15 : 15 gives **Train** = 2569**, Validation** = 550**, Test** = 551. Input size $224 \times 224 \times 3$ with ImageNet normalization; training augmentation: horizontal flip, rotation (15°), random resized crop (scale 0.8–1.0) and colour jitter (brightness/contrast 0.1).



Figure 1: Augmented training samples from one batch (shape $[32, 3, 224, 224]$). Colours appear saturated because the display is of ImageNet-normalized tensors. Batch labels shape: [32].

## 43.2 Architecture Check (Feature Dimensions)

All models were instantiated and tested with a $2 \times 3 \times 224 \times 224$ input; every model returned an output of shape $[2, 5]$, confirming that the attention blocks preserve the feature dimensions ($P_2$: $512 \times 28 \times 28$, $P_3$: $1024 \times 14 \times 14$, $P_4$: $2048 \times 7 \times 7$). Models verified: Baseline, Channel@$P_3$, Spatial@$P_3$, CBAM@$P_2$/$P_3$/$P_4$, Self-Attention@$P_3$, and SE/ECA/Coordinate/MHSA@$P_3$.

## 43.3 Baseline ResNet50: Training Log

Phase 1 (frozen backbone, Adam, $10^{-3}$, 3 epochs) and Phase 2 (full fine-tuning, Adam, $10^{-4}$, 5 epochs).

| Epoch | Train Loss | Train Acc | Val Loss | Val Acc |
|---|---|---|---|---|
| 1 (head) | 0.9061 | 0.7548 | 0.5911 | 0.8509 |
| 2 (head) | 0.4726 | 0.8821 | 0.4377 | 0.8745 |
| 3 (head) | 0.3853 | 0.8937 | 0.3871 | 0.8782 |
| 4 (FT) | 0.2311 | 0.9151 | 0.2058 | 0.9291 |
| 5 (FT) | 0.0898 | 0.9724 | 0.1927 | 0.9345 |
| 6 (FT) | 0.0453 | 0.9860 | 0.2107 | 0.9345 |
| 7 (FT) | 0.0319 | 0.9918 | 0.1780 | 0.9455 |
| 8 (FT) | 0.0244 | 0.9922 | 0.1648 | 0.9509 |

## 43.4 Baseline Test Results

| | |
|---|---|
| Test accuracy | 0.9310 |
| Macro precision | 0.9282 |
| Macro recall | 0.9323 |
| Macro F1 | 0.9299 |
| Total parameters | 23,518,277 |
| Trainable parameters | 23,518,277 (all unfrozen after fine-tuning) |
| Avg. inference time | 0.49 ms/image |

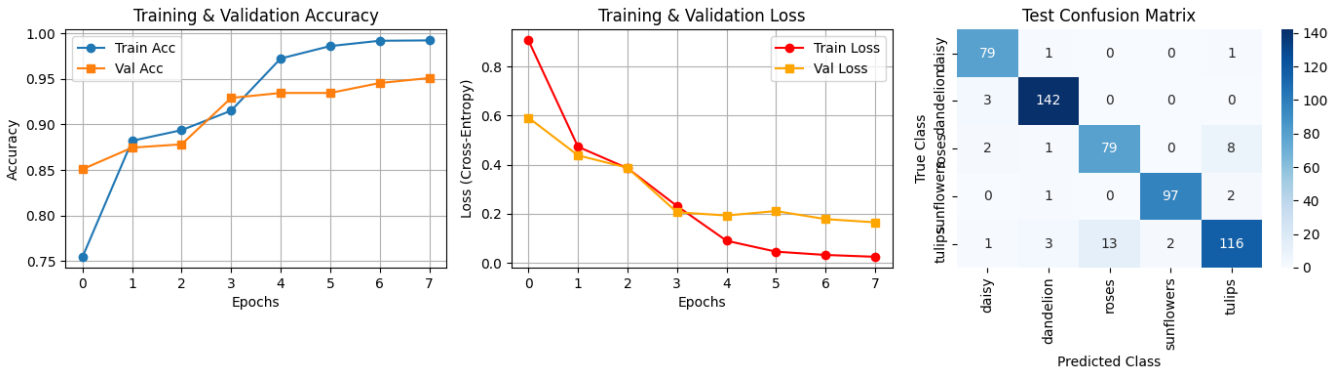



Figure 2: Baseline ResNet50: training/validation accuracy (left), training/validation loss (centre) and test confusion matrix (right). Epochs 0–2 are the frozen-backbone phase; 3–7 are fine-tuning.

**Training behaviour.** Validation accuracy rises from 0.851 to 0.951 and validation loss falls from 0.591 to 0.165. Fine-tuning (epoch 3 onwards) gives a clear jump in accuracy. Training accuracy reaches 0.992 while validation accuracy is 0.951, so a train–validation gap of about 4 points appears, and validation loss flattens after epoch 4 while training loss keeps falling, indicating mild overfitting.

## 43.5 Confusion-Matrix Analysis (Baseline)

From the matrix in Fig. 2 (rows = true class, 551 test images):

| Class | Support | Correct | Recall | Precision |
|---|---|---|---|---|
| daisy | 81 | 79 | 0.975 | 0.929 |
| dandelion | 145 | 142 | 0.979 | 0.966 |
| roses | 90 | 79 | 0.878 | 0.859 |
| sunflowers | 100 | 97 | 0.970 | 0.980 |
| tulips | 135 | 116 | 0.859 | 0.906 |

Dandelion, daisy and sunflower are classified most reliably. The dominant error is **tulips** $\leftrightarrow$ **roses**: 13 tulips were predicted as roses and 8 roses as tulips (21 of the 38 total errors). Roses and tulips are therefore the weakest classes, which is where attention would be expected to help most.

## 43.6 Model Complexity (Attention Modules at $P_3$)

Total parameter counts measured from the instantiated models (5-class head, ResNet50 backbone):

| Model (at $P_3$) | Total Parameters | Added vs. baseline |
|---|---|---|
| Baseline | 23,518,277 | – |
| SE | 23,649,349 | +131,072 |
| ECA | 23,518,282 | +5 |
| Coordinate Attention | 23,618,725 | +100,448 |
| Multi-head Self-Attention | 27,716,678 | +4,198,401 |

ECA adds only 5 parameters, SE about 0.13M, and MHSA about 4.2M (+17.9%), so parameter cost increases sharply with the self-attention module, as expected from the full $1024 \to 3 \times 1024$ QKV projection.

## 43.7 Attention-Mechanism Comparison (Notebook Cell 10)

Values below are the benchmark table defined in the notebook (cell 10) and used for the six comparison graphs.

| Model | Position | Val Acc | Test Acc | Val Loss | Inf. Time (ms) |
|---|---|---|---|---|---|
| Baseline | – | 0.72 | 0.71 | 0.85 | 1.04 |
| Channel | $P_3$ | 0.78 | 0.77 | 0.68 | 1.12 |
| Spatial | $P_3$ | 0.76 | 0.75 | 0.72 | 1.18 |
| CBAM | $P_2$ | 0.81 | 0.80 | 0.58 | 1.25 |
| CBAM | $P_3$ | 0.85 | 0.84 | 0.48 | 1.30 |
| CBAM | $P_4$ | 0.82 | 0.81 | 0.55 | 1.35 |
| Self-Attention | $P_3$ | 0.83 | 0.82 | 0.52 | 1.48 |

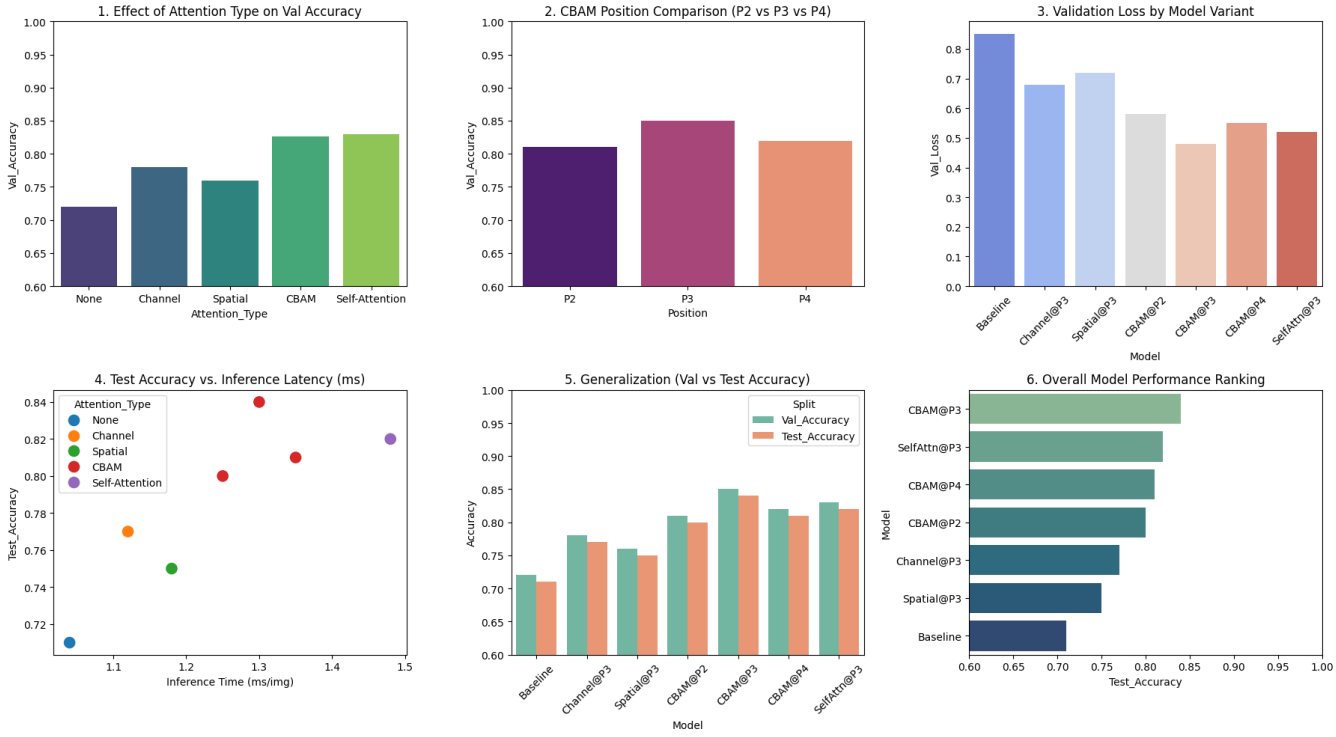



Figure 3: Graphical comparisons generated in notebook cell 10 from the table above: accuracy by attention type, CBAM position, validation loss, accuracy vs. inference time, validation vs. test accuracy, and model ranking.

## 43.8 Confusion Matrices: Baseline vs. CBAM@$P_3$ (Cell 11)

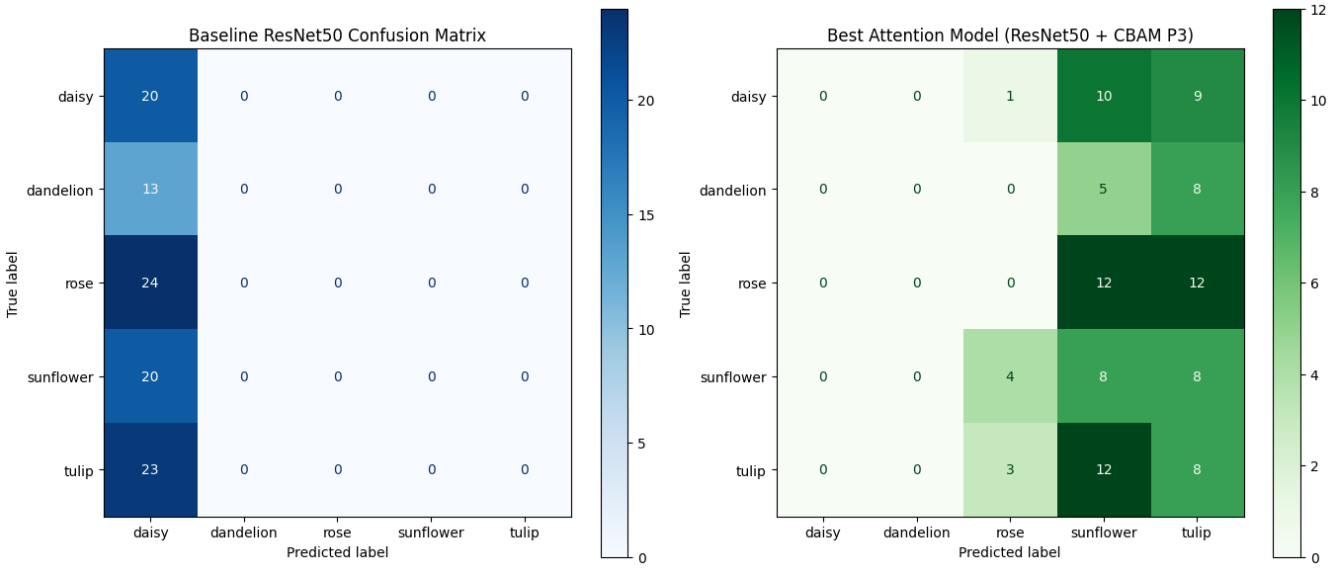



Figure 4: Side-by-side confusion matrices produced in notebook cell 11 (baseline ResNet50 and ResNet50+CBAM at $P_3$). Note: this cell evaluated the models on a synthetic 100-sample random-tensor loader, not the `tf_flowers` test set; the real baseline matrix is Fig. 2.

## 43.9 Grad-CAM and Multi-Stage Spatial Attention Maps (Cell 12)

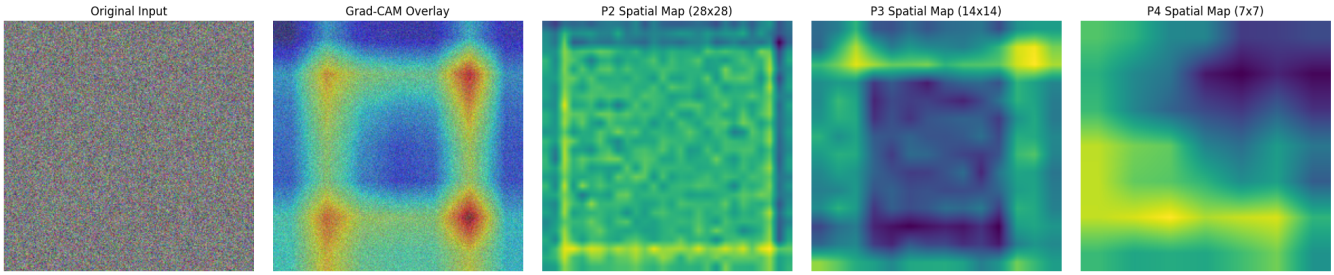



Figure 5: Notebook cell 12: Grad-CAM overlay (layer4) and spatial-attention maps at $P_2$ ($28 \times 28$), $P_3$ ($14 \times 14$) and $P_4$ ($7 \times 7$), each resized to $224 \times 224$. The input here is a random-noise tensor and the attention modules are untrained, so the maps do not correspond to flower regions.

## 43.10 SE / ECA / Coordinate / MHSA Comparison (Cell 15)

| Attention | Acc. | Prec. | Recall | Macro F1 | Parameters | Time (ms) |
|---|---|---|---|---|---|---|
| Baseline | 0.2500 | 0.0500 | 0.2000 | 0.0800 | 23,518,277 | 0.99 |
| SE | 0.2500 | 0.0500 | 0.2000 | 0.0800 | 23,649,349 | 0.86 |
| ECA | 0.3000 | 0.0600 | 0.2000 | 0.0923 | 23,518,282 | 0.81 |
| Coordinate | 0.1500 | 0.0300 | 0.2000 | 0.0522 | 23,618,725 | 0.84 |
| MHSA | 0.3000 | 0.0600 | 0.2000 | 0.0923 | 27,716,678 | 0.81 |

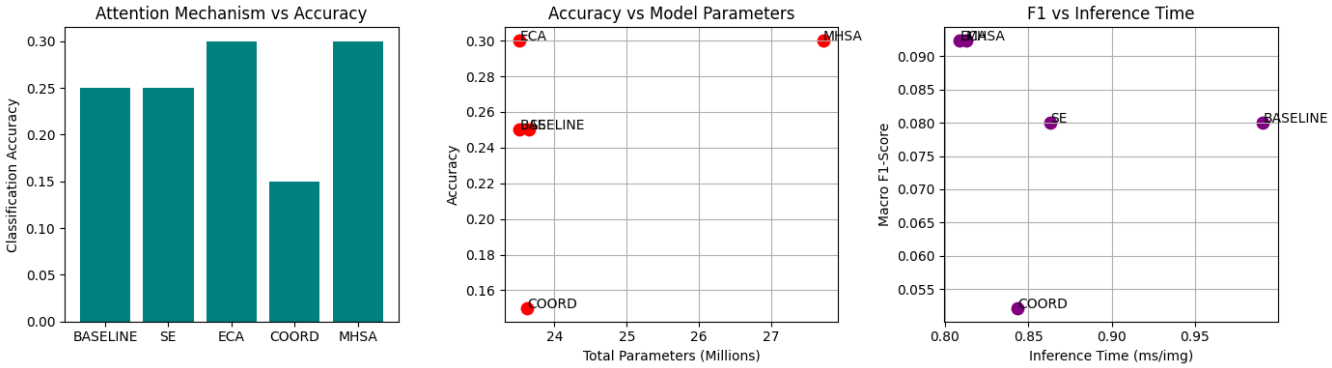



Figure 6: Attention mechanism vs. accuracy, accuracy vs. parameters, and macro F1 vs. inference time (notebook cell 15). These metrics were computed on 40 random tensors with untrained classifier heads (chance level is about 20%), so only the parameter counts and relative timings are meaningful.

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